

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start your trial today" section and create an account. Once completed the information select "Submit". It will redirect to the signup page.

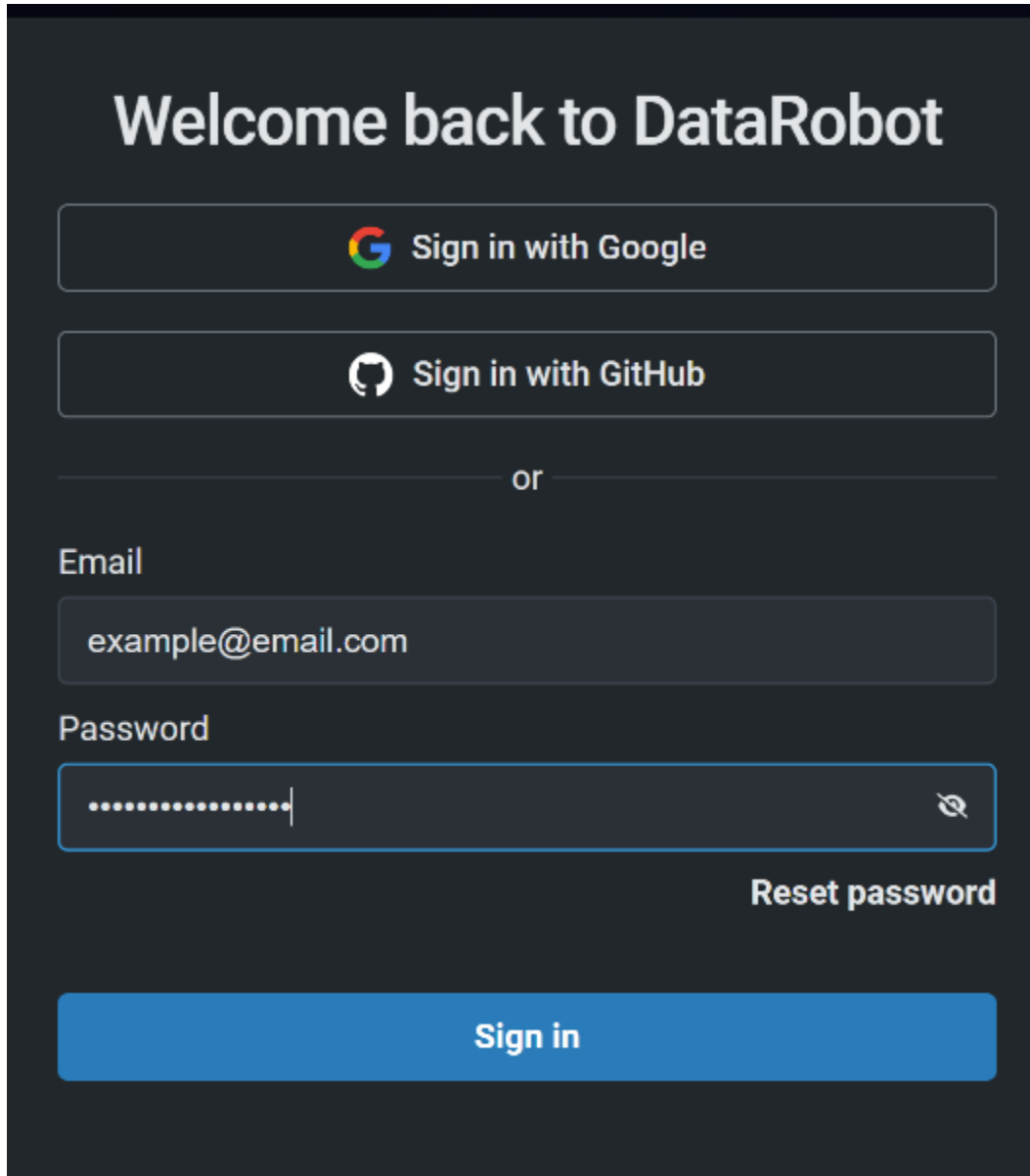
The screenshot shows the DataRobot website's homepage. At the top, there is a navigation bar with links for 'DataRobot', 'Why DataRobot', 'Apps & Agents', 'Platform', 'Resources', and 'About'. On the right side of the navigation bar are buttons for 'Log In', 'Try DataRobot', and 'Request a Demo'. Below the navigation bar is a large blue banner with the text 'Try DataRobot for free' and a subtext 'Experience how agents, applications, and an end-to-end suite of production-ready tools can accelerate your production use cases'. Below the banner, the page is divided into two main sections. On the left, there is a section titled 'Ship a production-grade agent, app, or PoC in 14 days' with a list of bullet points: 'Clone and build the app UI, underlying models, and production-ready components with a single line of code', 'Create a simple agentic workflow by securely connecting to systems of record and adding additional AI tools', and 'Customize all pre-built components based on your specific use case and data in DataRobot hosted notebooks or GUI'. Below this list is a dark box with the text 'Quickly deliver AI that impacts your business'. On the right, there is a section titled 'Start your trial today' which contains a form with the following fields: 'First Name: *' (with a placeholder 'Enter your first name'), 'Last Name: *' (with a placeholder 'Enter your last name'), 'Job Title: *' (with a placeholder 'Enter your job title'), 'Company: *' (with a placeholder 'Enter your company name'), and 'Email Address: *' (with a placeholder 'Enter your email address'). A plus sign icon is visible at the bottom right of the form.

Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 3: A new window will open; select the relevant option for signing in.

The image shows a login interface for DataRobot. At the top, it says "Welcome back to DataRobot". Below this are two buttons: "Sign in with Google" (with the Google logo) and "Sign in with GitHub" (with the GitHub logo). A horizontal line with the word "or" in the center separates these from the email/password section. The "Email" section has a text input field containing "example@email.com". The "Password" section has a text input field with masked characters (dots) and a toggle icon (an eye with a slash) on the right. Below the password field is a link that says "Reset password". At the bottom is a large blue button labeled "Sign in".

Welcome back to DataRobot

 Sign in with Google

 Sign in with GitHub

or

Email

example@email.com

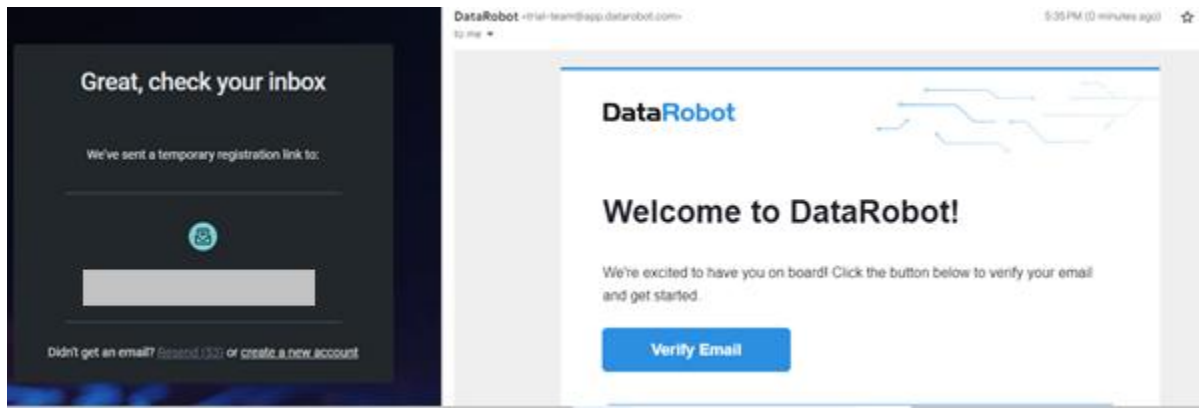
Password

.....| 

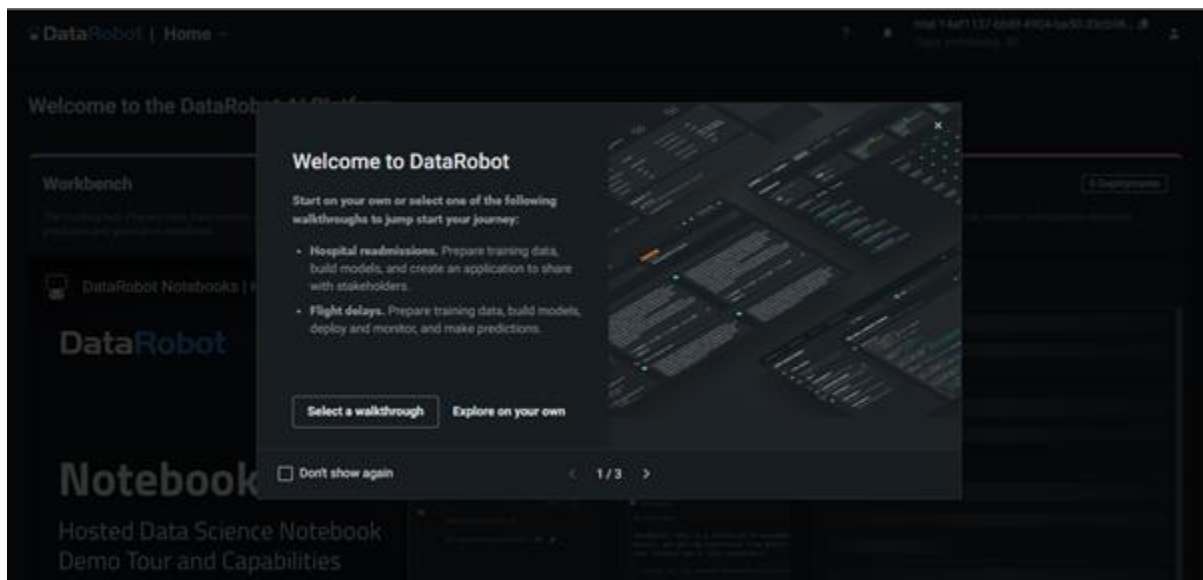
Reset password

Sign in

Step 4: Confirm your email by clicking **Verify Email** in your inbox.

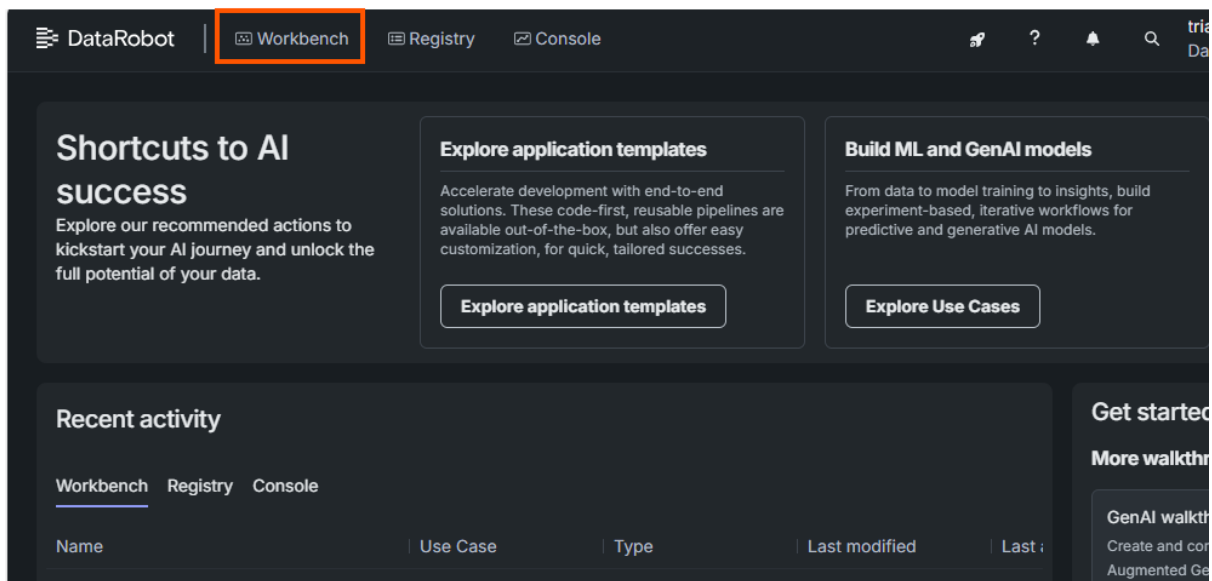


Step 5: Sign up and start your first experience of using the Generative AI tool.
 The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

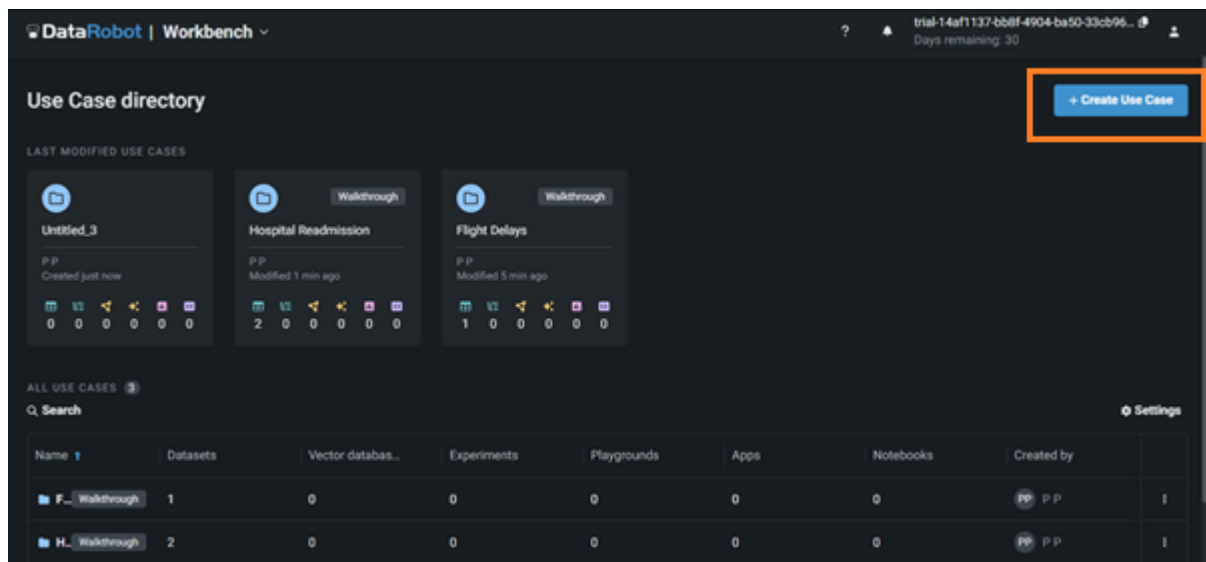


Task 2: Add a data set

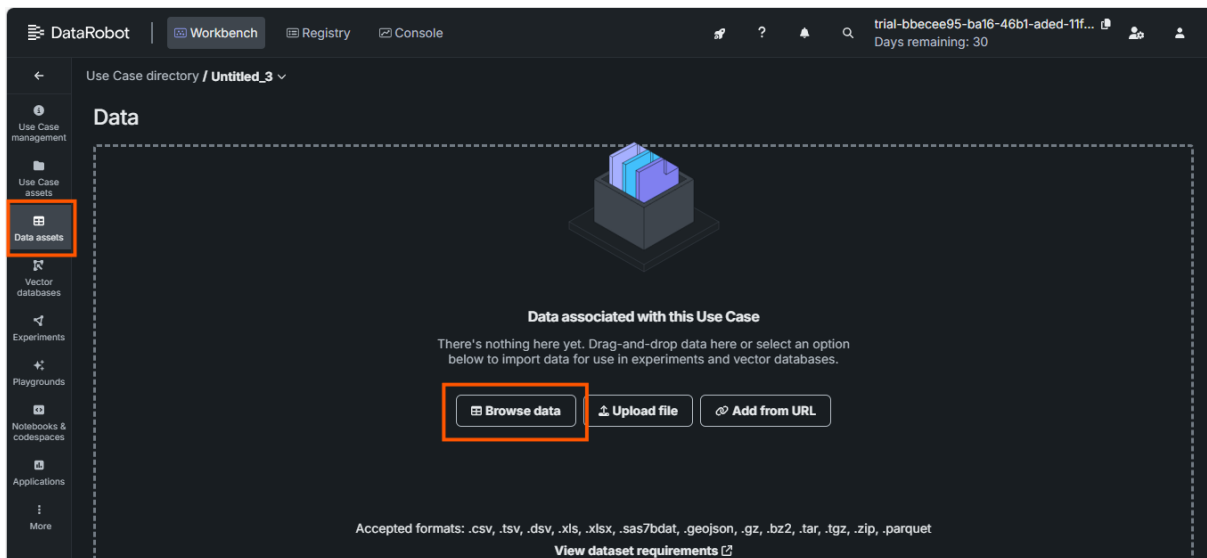
Step 8: The dashboard will appear shortly, and your screen will look as shown below.
 Click **Workbench**.



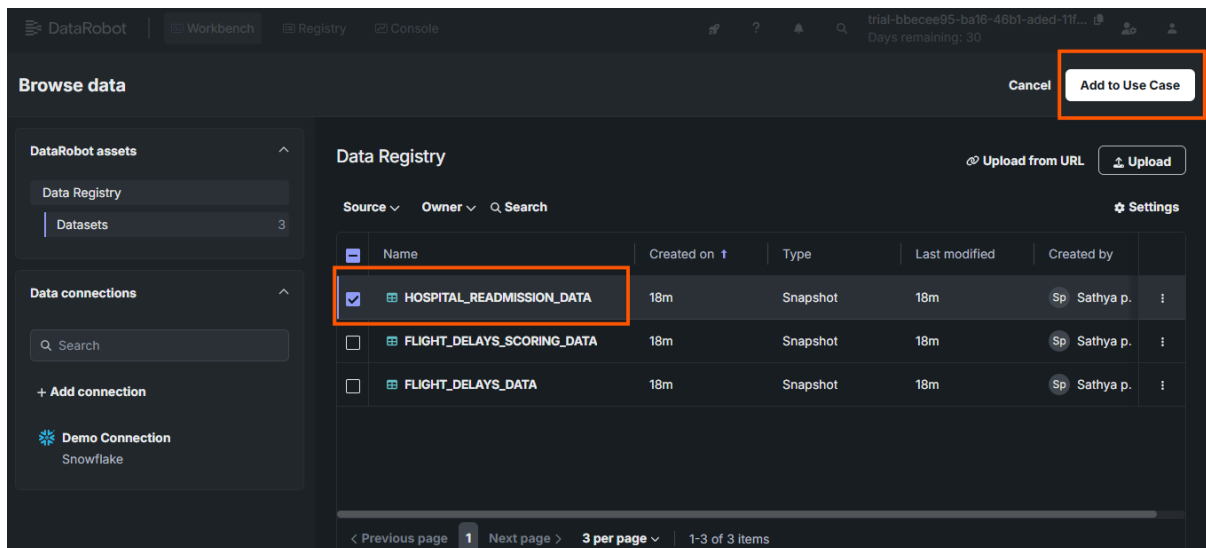
Step 9: Click **Create Use Case**.



Step 10: Click **Data assets** and select **Browse Data** to include the data set in your use case.



Step 10: You can select an in-built sample data set *HOSPITAL_READMISSION_DATA* and select **Add to Use Case**.



Step 11: Once you select the data set, you can see a preview of it by clicking on the dataset. You can also view the data set's features, as shown below. Click **Add to Use Case**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Add data

Close [Add to Use Case](#)

DATA STORES

[Upload](#) [+ Connect](#)

Q Search

- Data Registry DataRobot
- Demo Connection Snowflake

HOSPITAL_READMISSION_DATA

Data Registry / HOSPITAL_READMISSION_...

[Features](#) [Data preview](#)

race	gender	age	weight	admission_type_id	discharge_disposit
Categorical	Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Male	[50-60]		Emergency	Discharged to hom
Caucasian	Female	[70-80]		Urgent	Discharged/transfe
Caucasian	Male	[60-70]		Elective	
Caucasian	Male	[60-70]			Discharged to hom
Caucasian	Female	[40-50]		Not Available	Discharged to hom
Caucasian	Male	[90-100]		Emergency	Discharged/transfe
Caucasian	Male	[70-80]		Urgent	Discharged/transfe
Caucasian	Female	[80-90]		Urgent	Discharged to hom

[Snapshot sample](#) 51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Use Case directory

USE CASE

Untitled_4

[Add Data](#)

[All](#) [Data](#) [Vector databases](#) [Experiments](#) [Playgrounds](#) [Applications](#)

Type [Search](#) [Settings](#)

Name	Created By	Last Modified	Type	Source	Rows
HOSPITAL_READMISSION_DATA	PV Pratiksha V.	now	Snapshot	Snowflake	10000

< Previous 1 Next > 1 per page 1-1 of 1 items

Use Case info

DESCRIPTION
Add a description

CREATED
July 29, 2024

LAST MODIFIED
just now

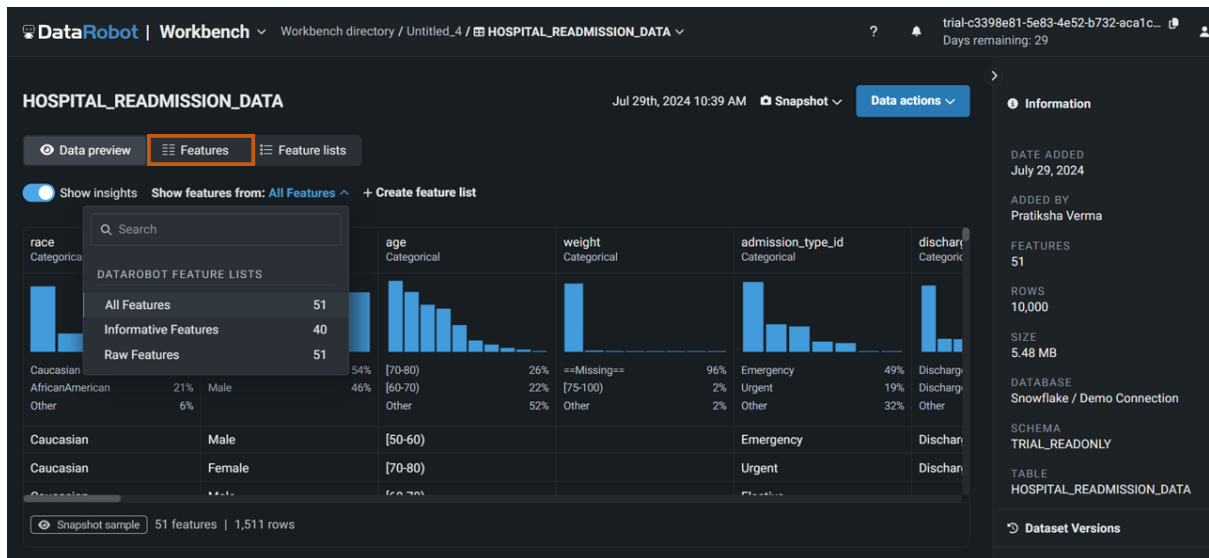
OWNER
PV Pratiksha Verma

TEAM
None

[Manage members](#)

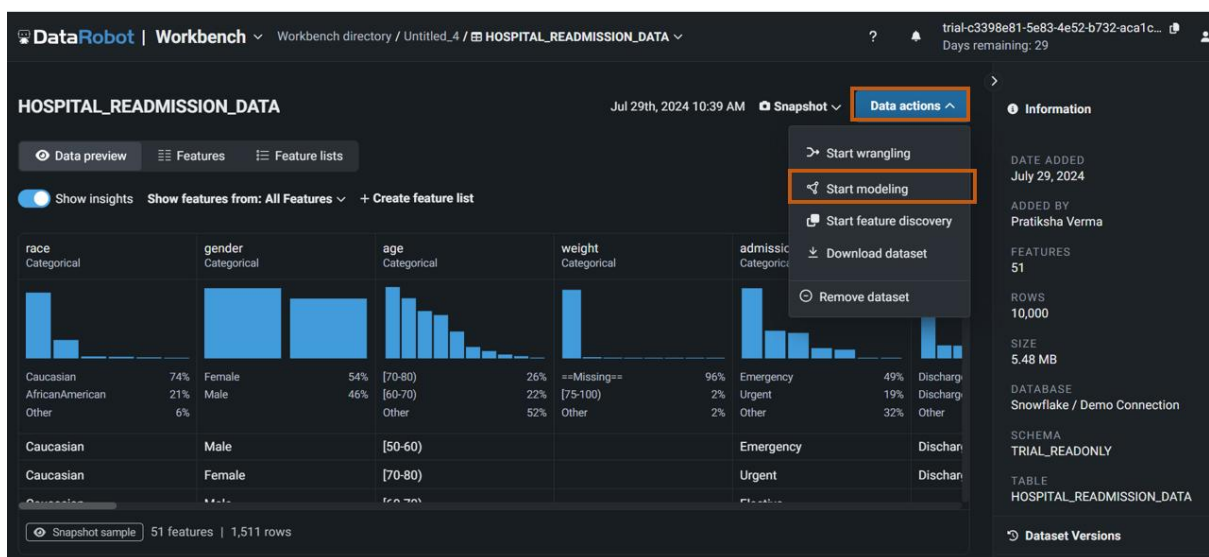
YOUR ROLE
Owner

Step 13: Explore the **All Features** menu to display specific features.



Task 3: Work on Data Modeling

Step 14: You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building under **Data actions** and select **Modelling**. It will take a while to prepare a data set for modelling.



Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Set up new experiment

Dataset Target Additional settings Exit < Back Next >

Target feature
Select the feature to make predictions on.

payer_code
pioglitazone
race
readmitted
readadlinide
troglitazone
tolbutamide
tolazamide
time_in_hospital

	Uniq...	Missi...	Mean	Std Dev
readmitted	7	9592	-	-
troglitazone	35	Categorical	1	0
tolbutamide	30	Categorical	2	0
tolazamide	36	Categorical	2	0
time_in_hospital	7	Numeric	14	0
			4.43	3.021

Experiment summary
HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset
Name: HOSPITAL_READMISSION_DATA
Rows: 10,000
Features: 51

Target
No target selected

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Set up new experiment

Dataset Target Additional settings Exit < Back **Next >**

Target feature
Select the feature to make predictions on.
readmitted

Target type: Binary classification ⓘ
Positive class: 0 1 1

Modeling mode
Set the mode used for selecting which blueprints to build when training models.
Quick Autopilot

Optimization metric
Set the metric used when training models to evaluate and optimize accuracy.
LogLoss (Accuracy) Recommended

Number of rows

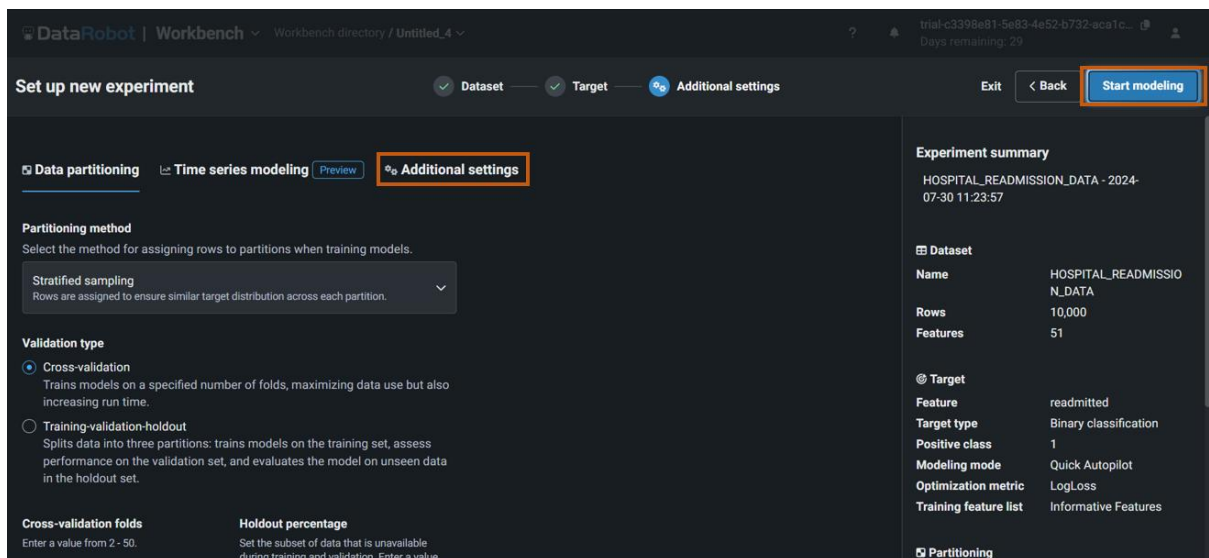
Experiment summary
HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset
Name: HOSPITAL_READMISSION_DATA
Rows: 10,000
Features: 51

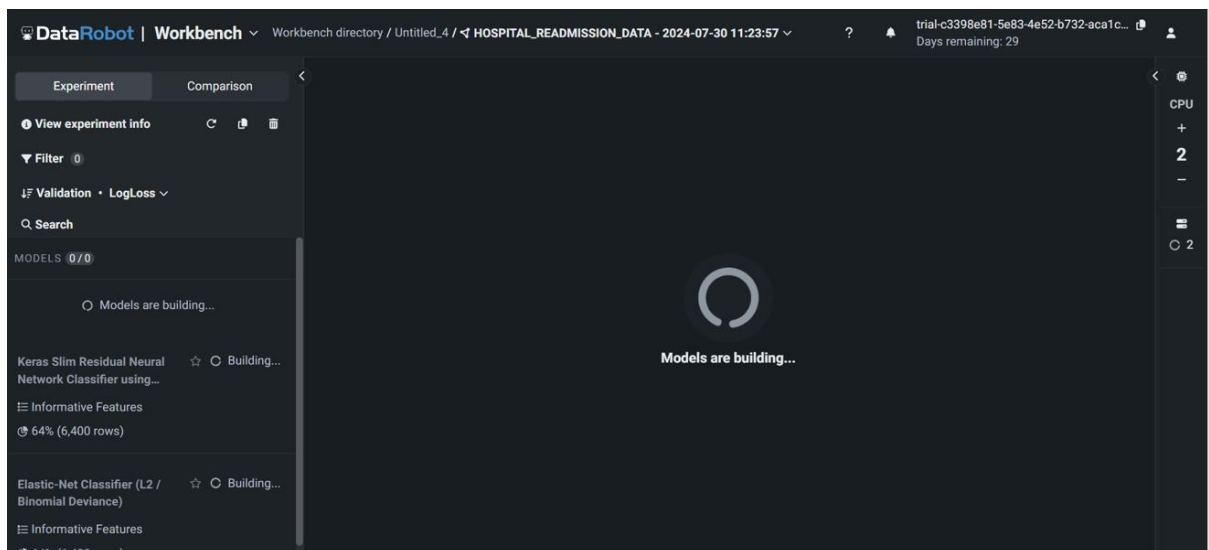
Target
Feature: readmitted
Target type: Binary classification
Positive class: 1
Modeling mode: Quick Autopilot
Optimization metric: LogLoss
Training feature list: Informative Features

Partitioning

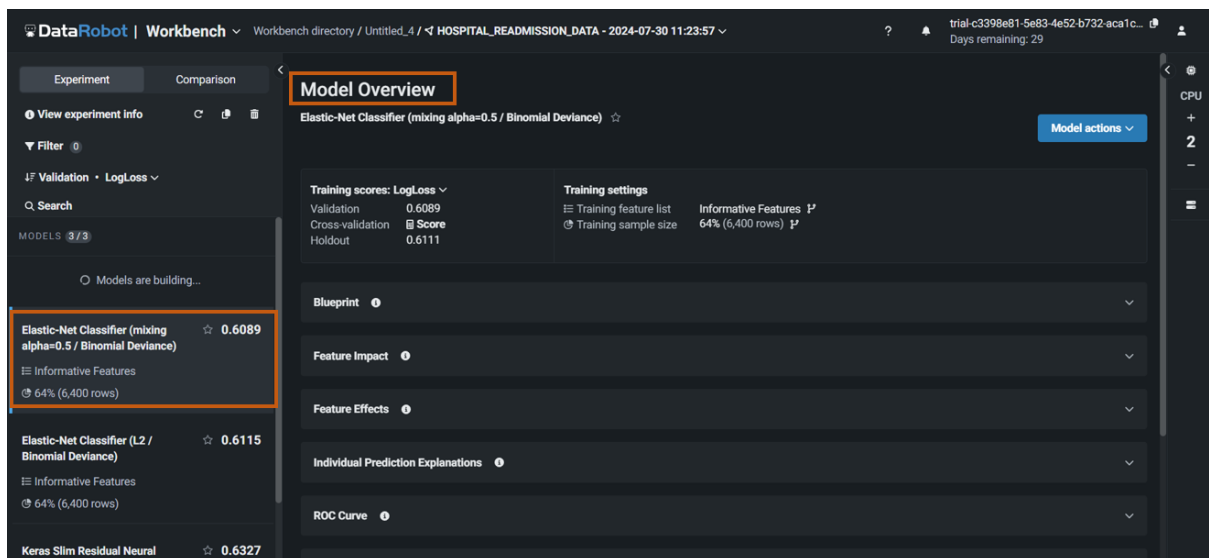
Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.



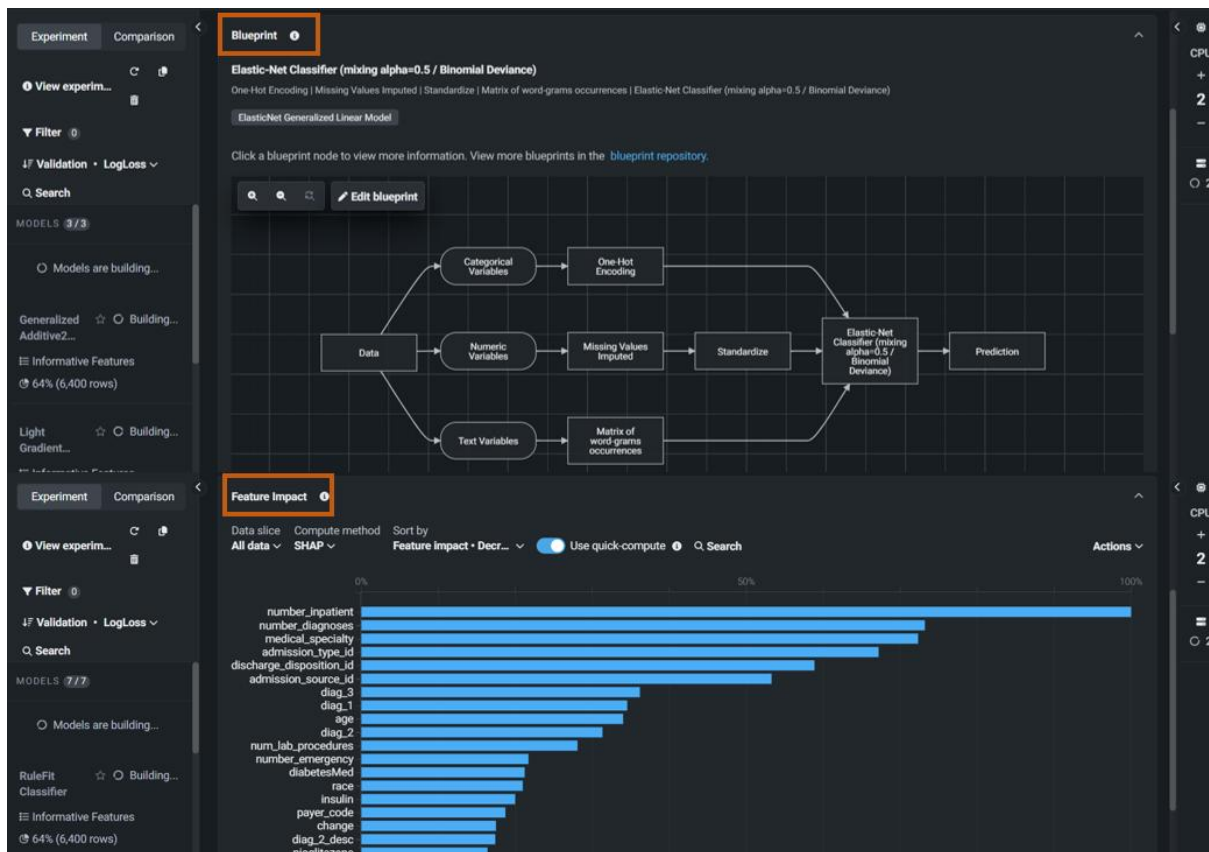
Step 18: Building models will take a while.



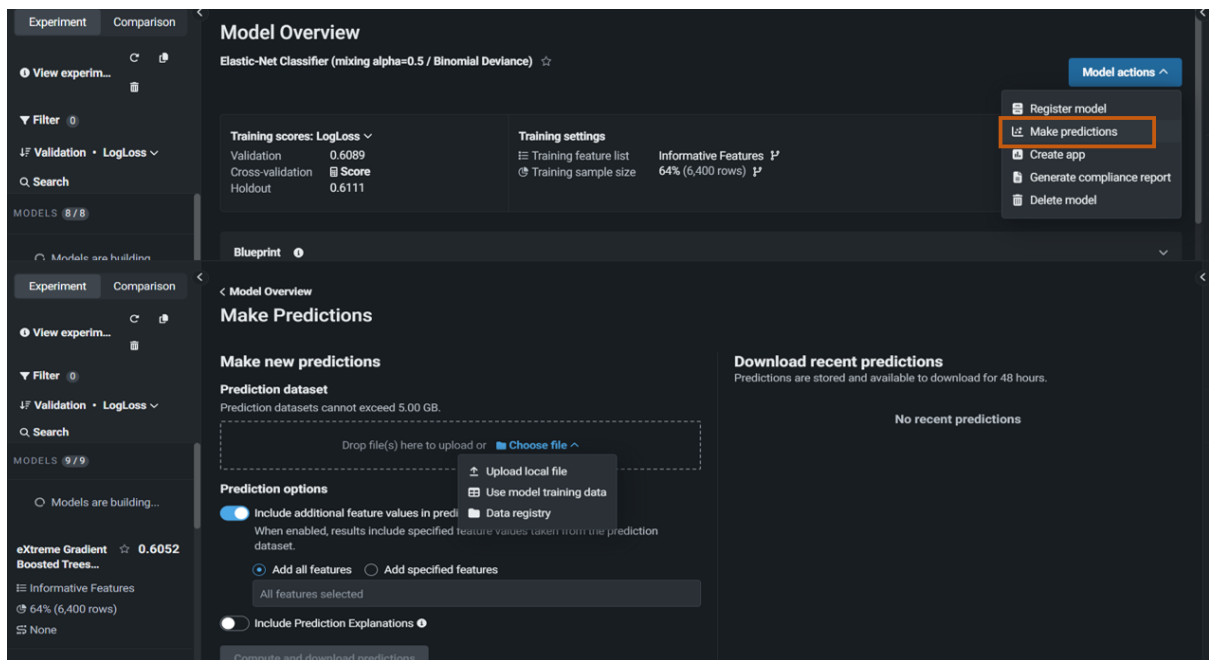
Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.



Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.



Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.



Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

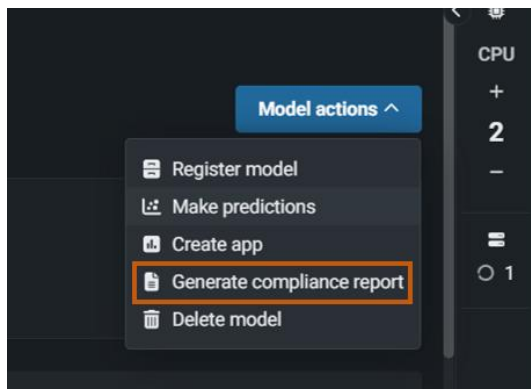


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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.